

PAPER_247: MUGE Merger Interaction Modulation — Tidal Gravity Boost with Exponential Decay

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — MUGEMergerInteractionModulationCalculator (Session 62, grok_share_8d951e12.txt 4th-pass) **Date:** March 2026 **Series:** Phase 2 Session 62 — §3.x Universal MUGE Sub-Term Integration

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot (1 - U_{bi}/F_U) \cdot e^{-\kappa t}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

Abstract

Galaxy mergers are the dominant channel for mass assembly at late cosmic times, temporarily amplifying tidal forces, star formation rates, and active galactic nuclei activity. This paper establishes the **merger interaction modulation sub-term** for MUGE, which captures the transient gravitational boost that occurs during and after a tidal encounter through an exponentially decaying interaction function $I(t) = I_0 \cdot \exp(-t/t_{\text{merger}})$.

The modulated gravity $g_{\text{merger}} = g_{\text{base}} \cdot (1 + I(t))$ peaks at $(1 + I_0) \sim 1.1$ times the base MUGE gravity at the moment of closest approach ($t = 0$) and relaxes exponentially to unity on the merger timescale t_{merger} . Two key characteristic times emerge: $t_{\text{half}} = t_{\text{merger}} \cdot \ln(2)$ (half-decay time) and $t_{\text{relax}} = t_{\text{merger}} \cdot \ln(I_0/0.01)$ (the epoch when the modulation drops below 1%).

The base gravity $g_{\text{base}} = (U_{g1} + U_{g4}) \cdot (1 + f_{\text{TRZ}})$ is itself built from the UQFF magnetic dipole term (U_{g1}), the vacuum-field correction ($U_{g4} = U_{g1} \cdot (1 - B/B_{\text{crit}})$), and the triadic resonance zone factor (f_{TRZ}). This term appears in the Antennae Galaxies and HUDF (Hubble Ultra-Deep Field) MUGE modules, confirming its astrophysical grounding in observed merger systems.

1. System Parameters and Equation Overview

Parameter	Symbol	Default Value	Units	Meaning
Peak boost amplitude	I_0	0.1	dimensionless	10% gravitational boost at $t=0$
Merger decay timescale	t_{merger}	$400 \text{ Myr} = 1.262 \times 10^{16} \text{ s}$	s	Exponential decay time
Body mass	M	$2 \times 10^{11} M_{\text{sun}}$	kg	Merging galaxy mass
Separation radius	r	30 kly	m	Tidal interaction scale

Parameter	Symbol	Default Value	Units	Meaning
TRZ factor	f_TRZ	0.1	dimensionless	Triadic resonance zone contribution
Critical B field	B_crit	4.4×10^{13}	T	Magnetar QED critical field

Primary equations:

$$I(t) = I_0 \cdot \exp(-t / t_{\text{merger}})$$

$$U_{g1} = G \cdot M / r^2 \quad [\text{magnetic dipole gravity}]$$

$$U_{g4} = U_{g1} \cdot (1 - B / B_{\text{crit}}) \quad [\text{vacuum-field correction}]$$

$$g_{\text{base}} = (U_{g1} + U_{g4}) \cdot (1 + f_{\text{TRZ}})$$

$$g_{\text{merger}} = g_{\text{base}} \cdot (1 + I(t)) \quad [\text{modulated merger gravity}]$$

Characteristic times:

$$t_{\text{half}} = t_{\text{merger}} \cdot \ln(2) \quad \sim 277 \text{ Myr}$$

$$t_{\text{relax}} = t_{\text{merger}} \cdot \ln(I_0 / 0.01) \quad \sim 920 \text{ Myr} \quad (I \text{ drops below } 1\%)$$

2. Core Physics Derivation

2.1 Exponential Decay of Tidal Interaction

During a galaxy merger, the tidal force is dominated by the time of closest approach. After periapsis, the two galaxies recede, and the gravitational perturbation decays. The simplest physically motivated model is exponential decay with a characteristic timescale t_{merger} — the tidal interaction time, roughly proportional to the orbital period at the merger separation.

$$I(t) = I_0 \cdot e^{-t/t_{\text{merger}}}$$

At $t = 0$ (closest approach): $I = I_0 = 0.1$, giving a 10% boost. As $t \gg 8 \cdot t_{\text{merger}}$: $I \approx 0$, recovering unperturbed base gravity.

This parametrisation is consistent with N-body merger simulations (e.g., Springel & Hernquist 2005) which find that star formation rate enhancements decay exponentially with timescales of 200–600 Myr for major mergers.

2.2 Base Gravity Construction from UQFF Sub-Terms

The merger modulation amplifies the UQFF base gravity, not the Newtonian gravity alone. The base gravity is:

$$U_{g1} = G \cdot M / r^2 \quad [\text{Newtonian-equivalent dipole term}]$$

$$U_{g4} = U_{g1} \cdot (1 - B / B_{\text{crit}}) \quad [\text{vacuum-field reduction: } U_{g4} < U_{g1} \text{ for } B > 0]$$

$$\begin{aligned} g_{\text{base}} &= (U_{g1} + U_{g4}) \cdot (1 + f_{\text{TRZ}}) \\ &= U_{g1} \cdot (2 - B / B_{\text{crit}}) \cdot (1 + f_{\text{TRZ}}) \end{aligned}$$

For $B \ll B_{\text{crit}}$ (galactic fields $\sim 10^{10}$ T): $U_{g4} \sim U_{g1}$, so $g_{\text{base}} \sim 2 \cdot U_{g1} \cdot (1 + f_{\text{TRZ}})$. The TRZ factor $f_{\text{TRZ}} = 0.1$ adds a 10% triadic resonance contribution, giving $g_{\text{base}} \sim 2.2 \cdot U_{g1}$.

Peak modulated gravity:

$$\begin{aligned} g_{\text{merger}}(t=0) &= g_{\text{base}} \cdot (1 + I_0) \\ &= 2.2 \cdot U_{g1} \cdot 1.1 \\ &\sim 2.42 \cdot G \cdot M / r^2 \end{aligned}$$

This $\sim 2.4 \times$ Newtonian gravity at closest approach — consistent with observed tidal distortion amplitudes in Antennae-class mergers.

2.3 Temporal Decay Analysis

Half-life: $t_{\text{half}} = t \cdot \ln(2)$. For $t = 400$ Myr: $t_{\text{half}} \sim 277$ Myr. Half the initial merger boost is dissipated in ~ 277 Myr — the timeframe over which the Antennae system's starburst peaks and begins to fade.

1% relaxation: $t_{\text{relax}} = t \cdot \ln(I_0/0.01) = 400 \cdot \ln(10) \sim 920$ Myr. The galaxy pair is effectively unperturbed after ~ 1 Gyr, consistent with the dynamical friction timescale for major mergers.

Instantaneous merger rate: $dI/dt = -I_0/t \cdot \exp(-t/t)$ — most rapid change at $t = 0$ (peak merger), slowing as the system relaxes.

2.4 HUDF Application

In the Hubble Ultra-Deep Field modules, this term models the cumulative effect of merger-induced gravity boosts across a population of galaxies at $z \sim 1-6$. The average boost g_{merger} over the merger population is:

$$g_{\text{merger}} = g_{\text{base}} \cdot (1 + I_0 \cdot t / T_{\text{observe}})$$

where T_{observe} is the observation window. For the HUDF ($T \sim 13$ Gyr), the contribution is small but non-zero — merger-driven gravity remains a detectable perturbation in the deep field.

3. Exponential Relaxation Theorem

Theorem (MUGE Merger Relaxation): For any merger with initial boost I_0 and timescale t_{merger} , the modulated gravity converges to base MUGE gravity exponentially: $g_{\text{merger}}(t) \rightarrow g_{\text{base}}$ as $t \rightarrow \infty$. The total integrated boost is:

$$\int_0^\infty [g_{\text{merger}}(t) - g_{\text{base}}] dt = g_{\text{base}} \cdot I_0 \cdot t_{\text{merger}}$$

For the default Antennae parameters: integrated boost $\sim g_{\text{base}} \times 0.1 \times 400$ Myr = 40 Myr $\cdot g_{\text{base}}$. This is the total additional gravitational impulse delivered to the merging system — a directly observable quantity through the system's orbital energy deficit.

4. Observational Predictions / Validation

- **Antennae Galaxies (NGC 4038/4039):** $t_{\text{merger}} \sim 400$ Myr matches the observed post-periapsis age of ~ 400 Myr; the current boost $I(400 \text{ Myr}) \sim I_0/e \sim 0.037$ — a 3.7% gravity enhancement detectable in velocity dispersion measurements.

- **HUDF merger fraction:** At $z \sim 1$, the HUDF merger fraction is $\sim 30\%$; the modulation term predicts an average 10% boost to the gravity of the merger sub-population, measurable as a 10% excess in stellar velocity dispersions for interacting pairs vs. isolated galaxies.
 - **Post-merger quiescence:** $t_{\text{relax}} \sim 920$ Myr predicts AGN feedback cessation timescale — consistent with observed AGN lifetimes in post-merger hosts (0.5–1 Gyr; Schawinski et al. 2015).
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5. References

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